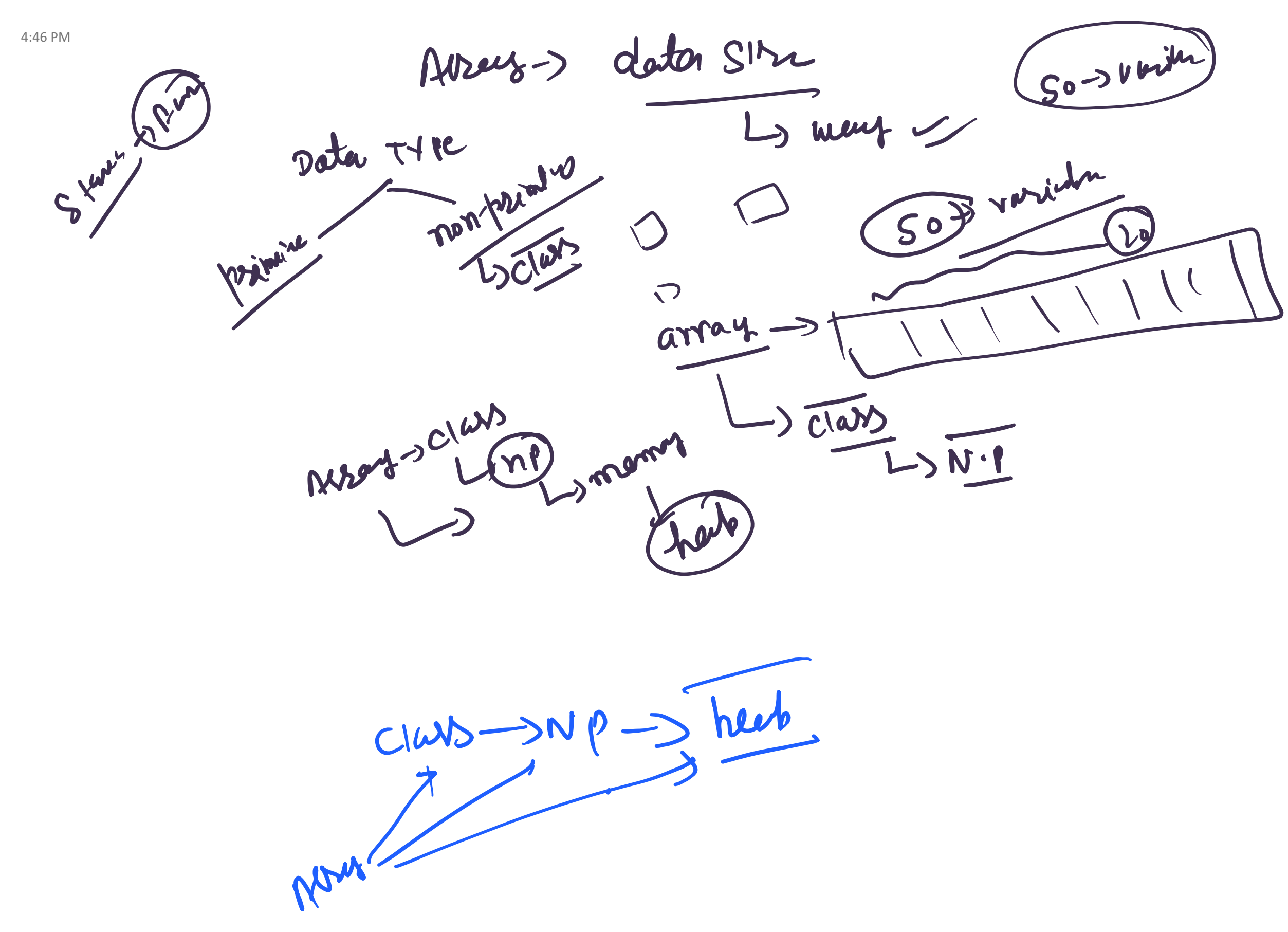
Lec-7

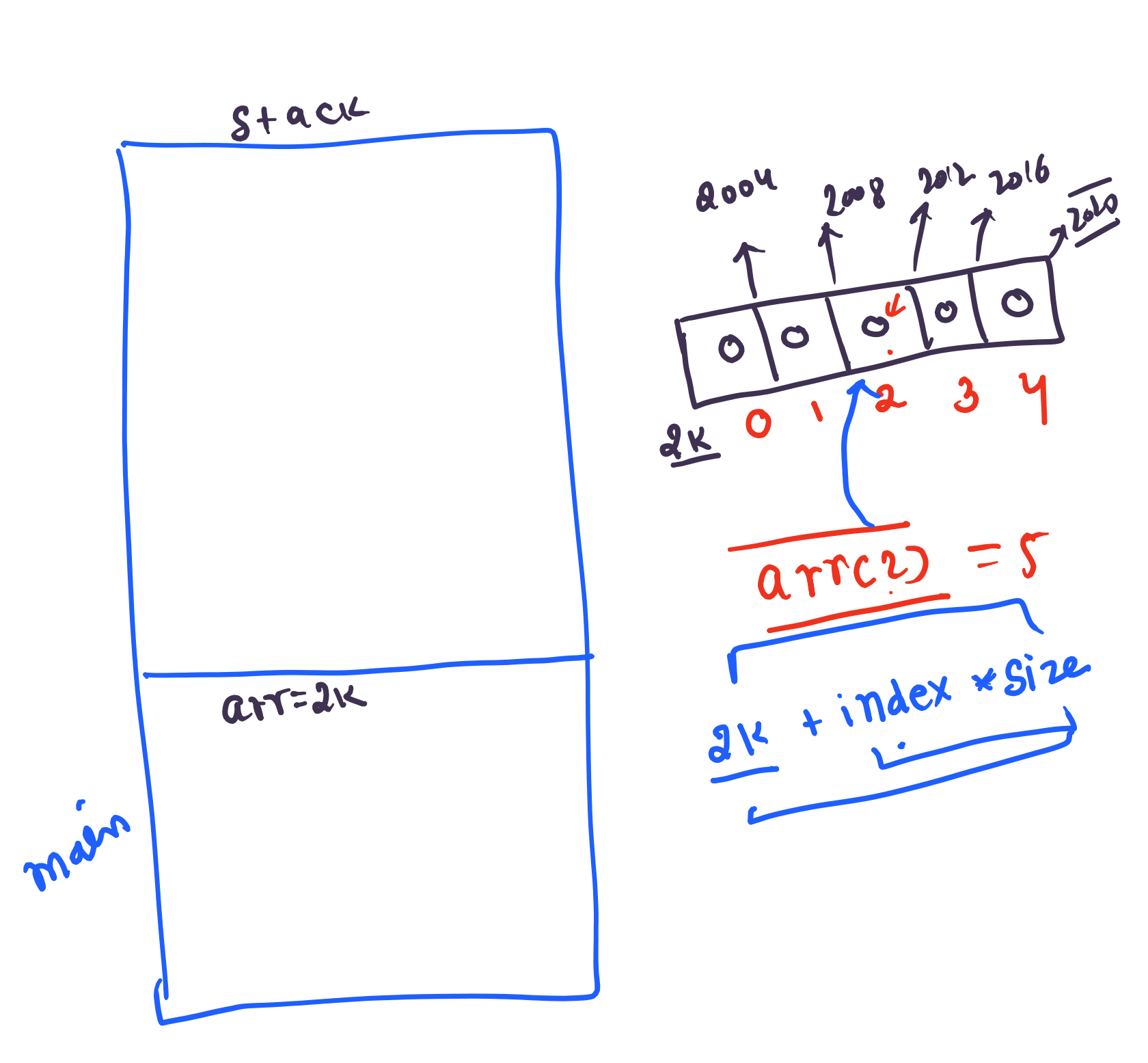
06/09/26 4:46 PM



Arrays -> data Size
L> many
So -> variable
Stack / Print
Data Type
primitive
non primitive
L> class
array ->
50 variable
20
L> class
L> N.P
Array -> class
L> NP
L> memory
heab
class -> NP -> heab
Array

```
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int[] arr = new int[5];
	System.out.println(arr);
}
```

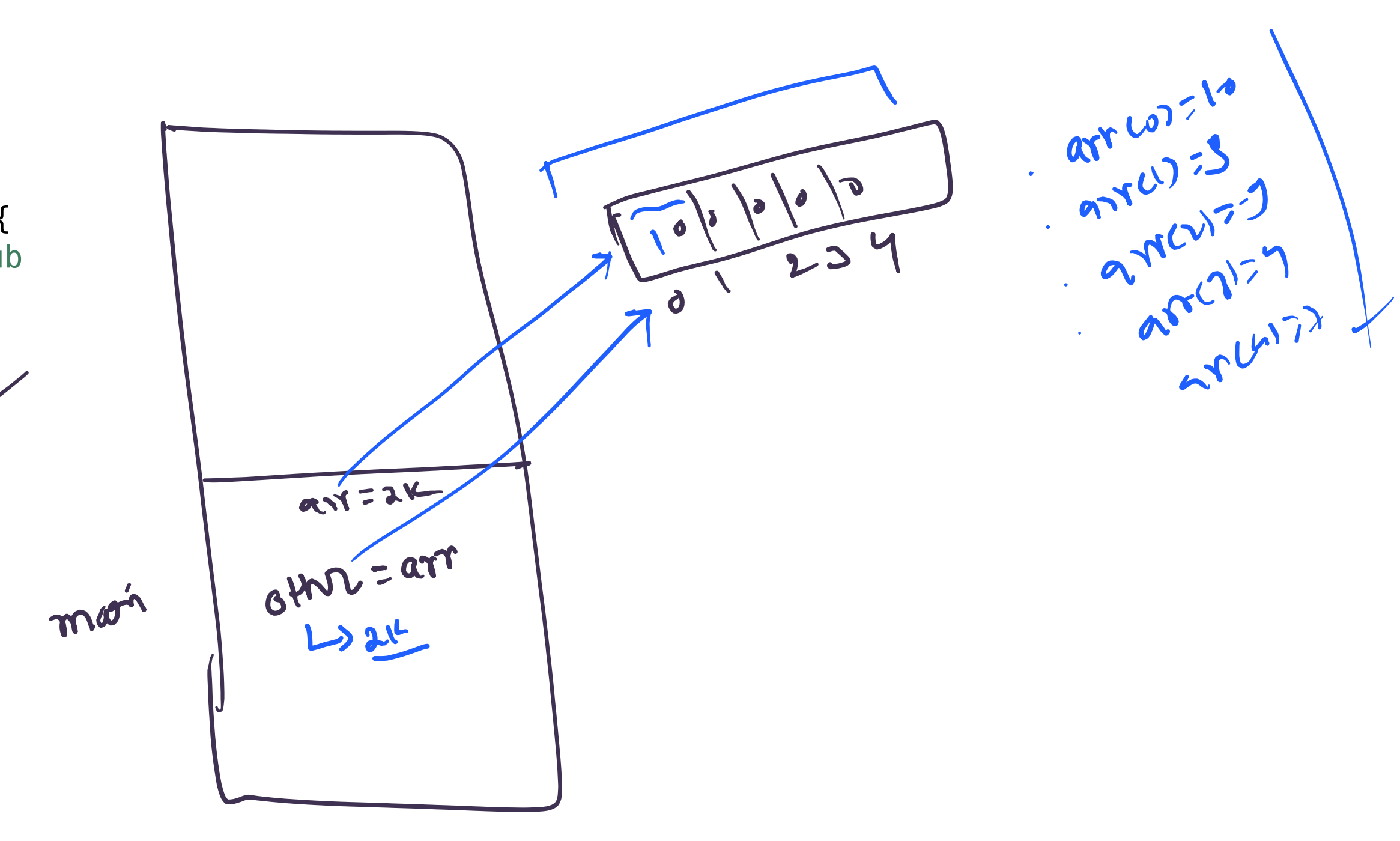
① why zero base Index ?



stack
arr=2K
main
2004
2008
2012
2016
2020
0
0
0
0
0
2K
0 1 2 3 4
arr[2] = 5
2K + index * Size

```
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int a = 9;
	int[] arr = new int[5];
	System.out.println(arr);
	System.out.println(arr.length);
	int[] other = arr;
}
```

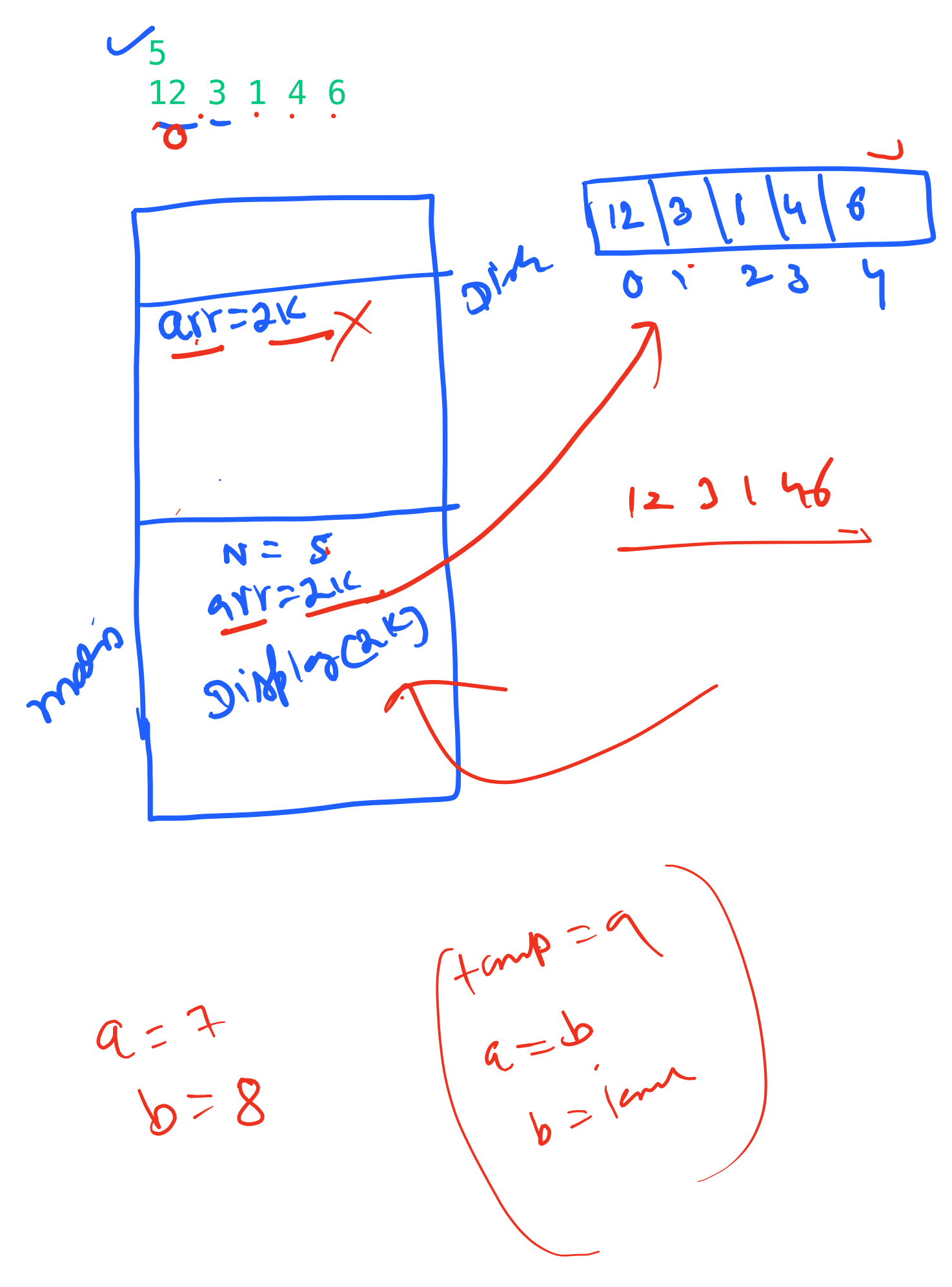
Other(0)



main
arr=2K
other=arr
L> 2K
10 3 2 9 7
0 1 2 3 4
arr[0]=10
arr[1]=3
arr[2]=2
arr[3]=9
arr[4]=7

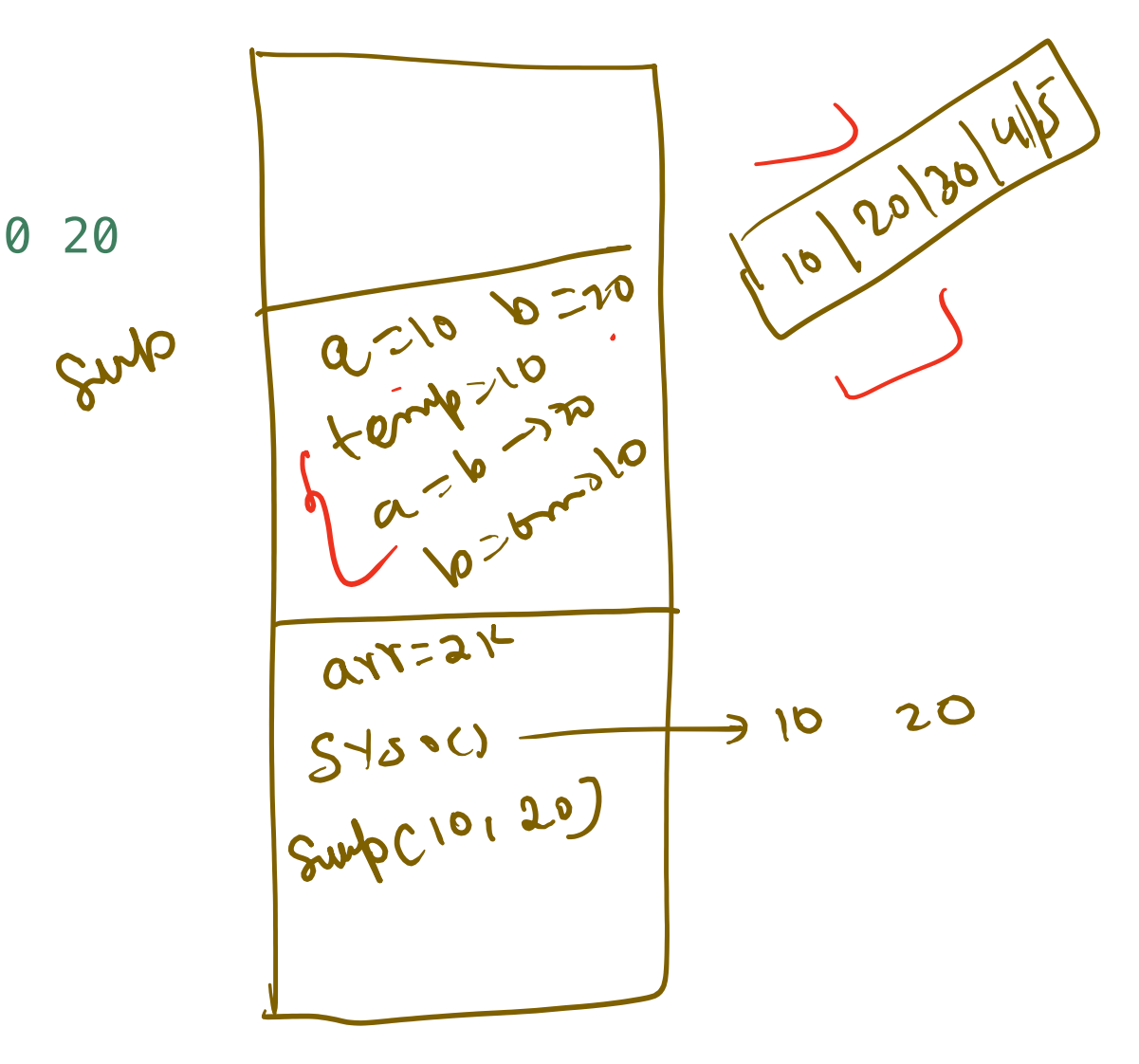
```
import java.util.Scanner;

public class User_Input_Output {
	public static void main(String[] args) {
		// TODO Auto-generated method stub
		Scanner sc = new Scanner(System.in);
		int n = sc.nextInt();
		int[] arr = new int[n];
		for (int i = 0; i < arr.length; i++) {
			arr[i]=sc.nextInt();
		}
		Display(arr);
	}
	public static void Display(int []arr) {
		for (int i = 0; i < arr.length; i++) {
			System.out.print(arr[i]+" ");
		}
		System.out.println();
	}
}
```

L> 2K



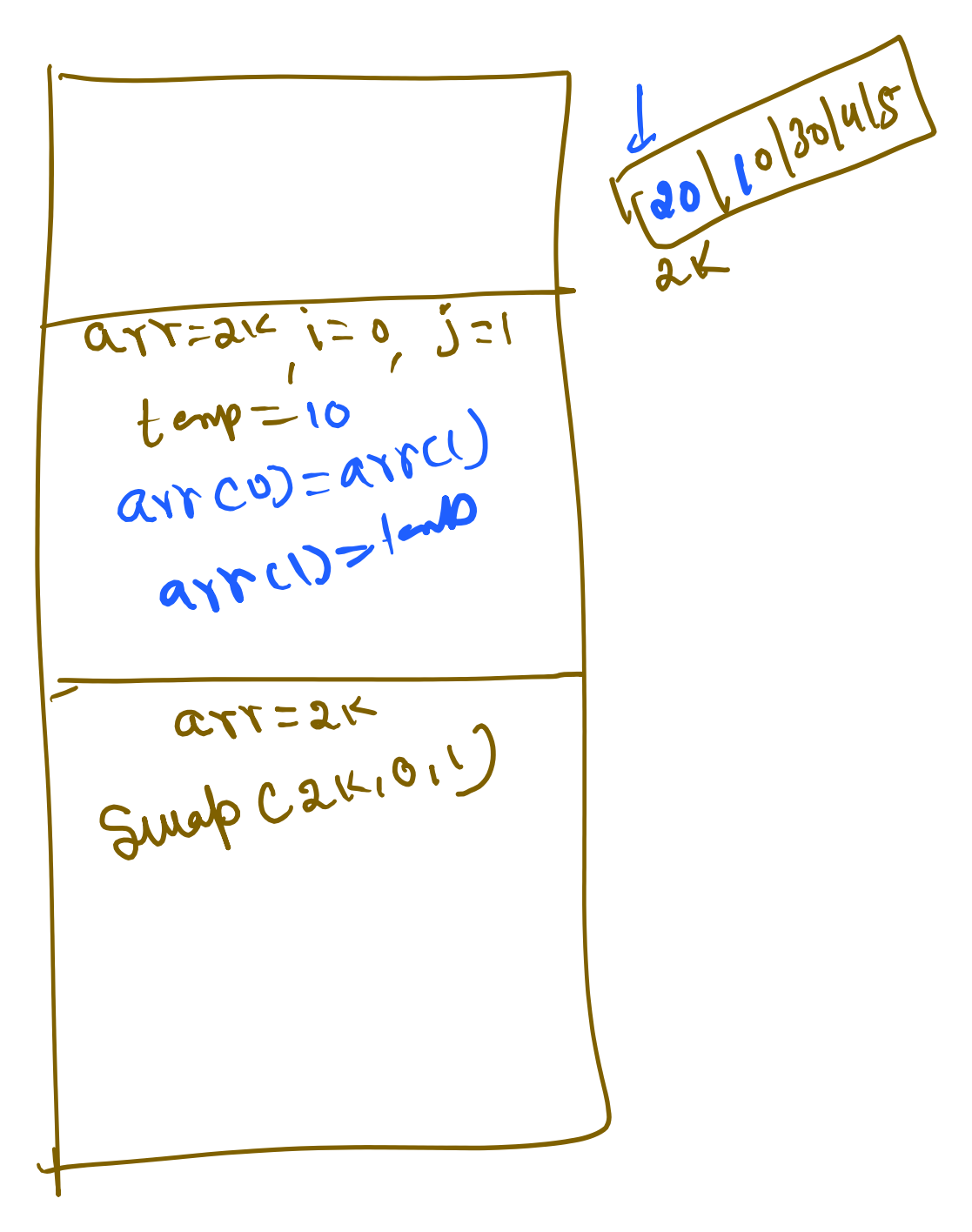
5
12 3 1 4 6
arr=2K
Display
N = 5
arr=2K
Display(2K)
main
12 3 1 4 6
0 1 2 3 4
12 3 1 4 6
a=7
b=8
temp=a
a=b
b=temp

```
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int[] arr = { 10, 20, 30, 4, 5 };
//	int[] arr1 = new int []{ 10, 20, 30, 4, 5 };
	System.out.println(arr[0] + " " + arr[1]);// 10 20
	swap(arr[0], arr[1]);
	System.out.println(arr[0] + " " + arr[1]);
}

public static void swap(int a, int b) {
	int temp = a;
	a = b;
	b = temp;
}
```



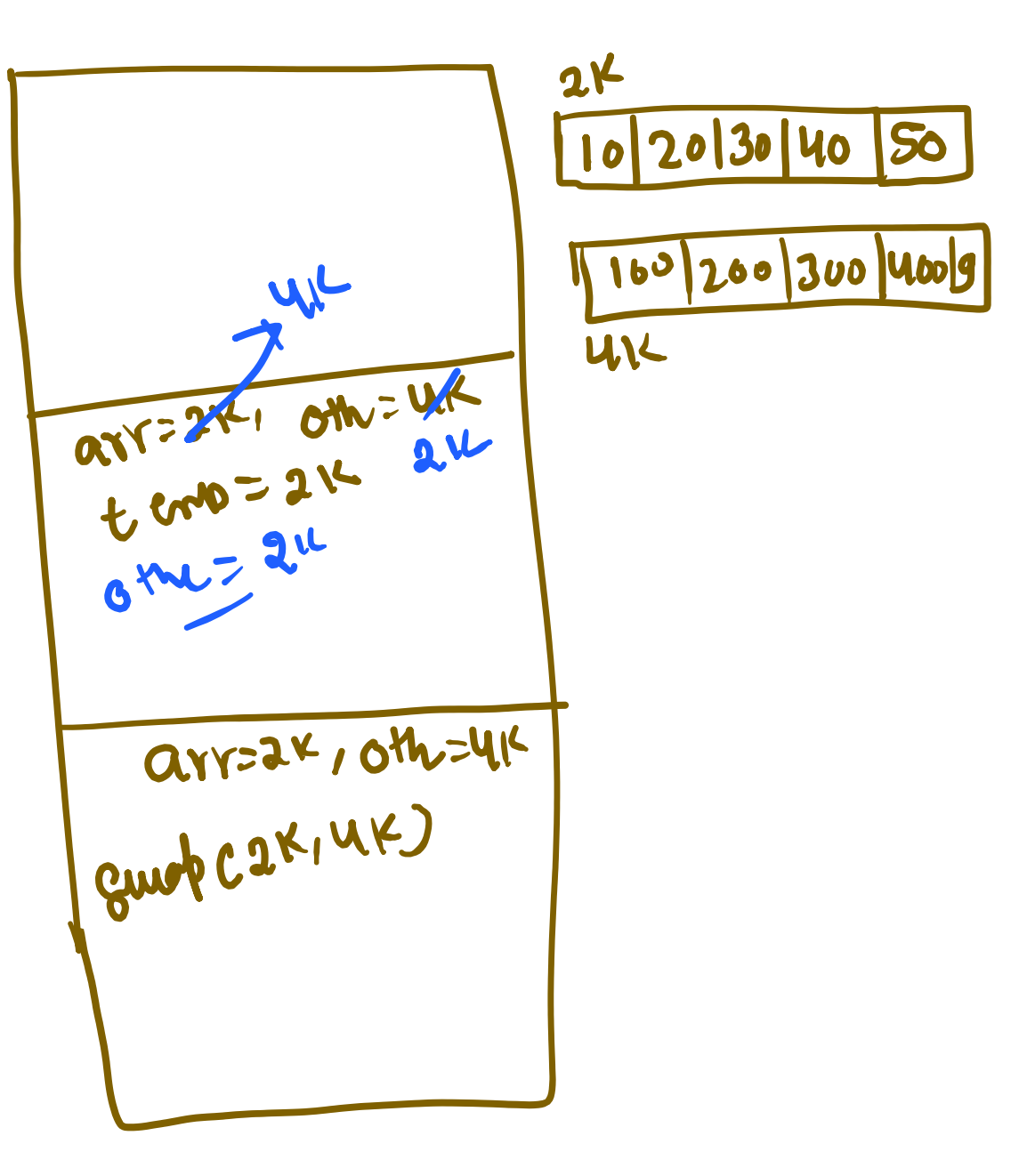
Swap
a=10 b=20
temp=10
a=b->20
b=temp->10
arr=2K
Sys.o()
Swap(10, 20)
-> 10 20
10 20 30 4 5

```
public static void main(String[] args) {
	int[] arr = { 10, 20, 30, 4, 5 };
	System.out.println(arr[0] + " " + arr[1]);// 10 20
	swap(arr, 0, 1);
	System.out.println(arr[0] + " " + arr[1]);
}

public static void swap(int[] arr, int i, int j) {
	int temp = arr[i];
	arr[i] = arr[j];
	arr[j] = temp;
}
```



arr=2K i=0 j=1
temp=10
arr[0]=arr[1]
arr[1]=temp
arr=2K
Swap(2K, 0, 1)
20 10 30 4 5
2K

```
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int[] arr = { 10, 20, 30, 40, 50 };
	int[] other = { 100, 200, 300, 400, 9 };
	System.out.println(arr[0] + " " + other[0]);// 10 100
	Swap(arr, other);
	System.out.println(arr[0] + " " + other[0]);//
}

public static void Swap(int[] arr, int[] other) {
	// TODO Auto-generated method stub
	int[] temp = arr;
	arr = other;
	other = temp;
}
```



2K
10 20 30 40 50
100 200 300 400 9
4K
arr=2K, oth=4K
temp=2K
other=2K
4K
2K
arr=2K, oth=4K
Swap(2K, 4K)

```
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int[] arr = { 3, 5, 6, 2, 4, 16, 7, 8, 9 };
	int item = 5;
}

public static int Search(int[] arr, int item) {

}
```

item=5

[2, 4, 5, 7, 8]

```
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int[] arr = { 3, 5, 6, 2, 4, 16, 7, 8, 9 };
}

public static int Maximum(int[] arr) {

}
```

-∞ 8 6 4

max = ~~arr[0]~~ -∞

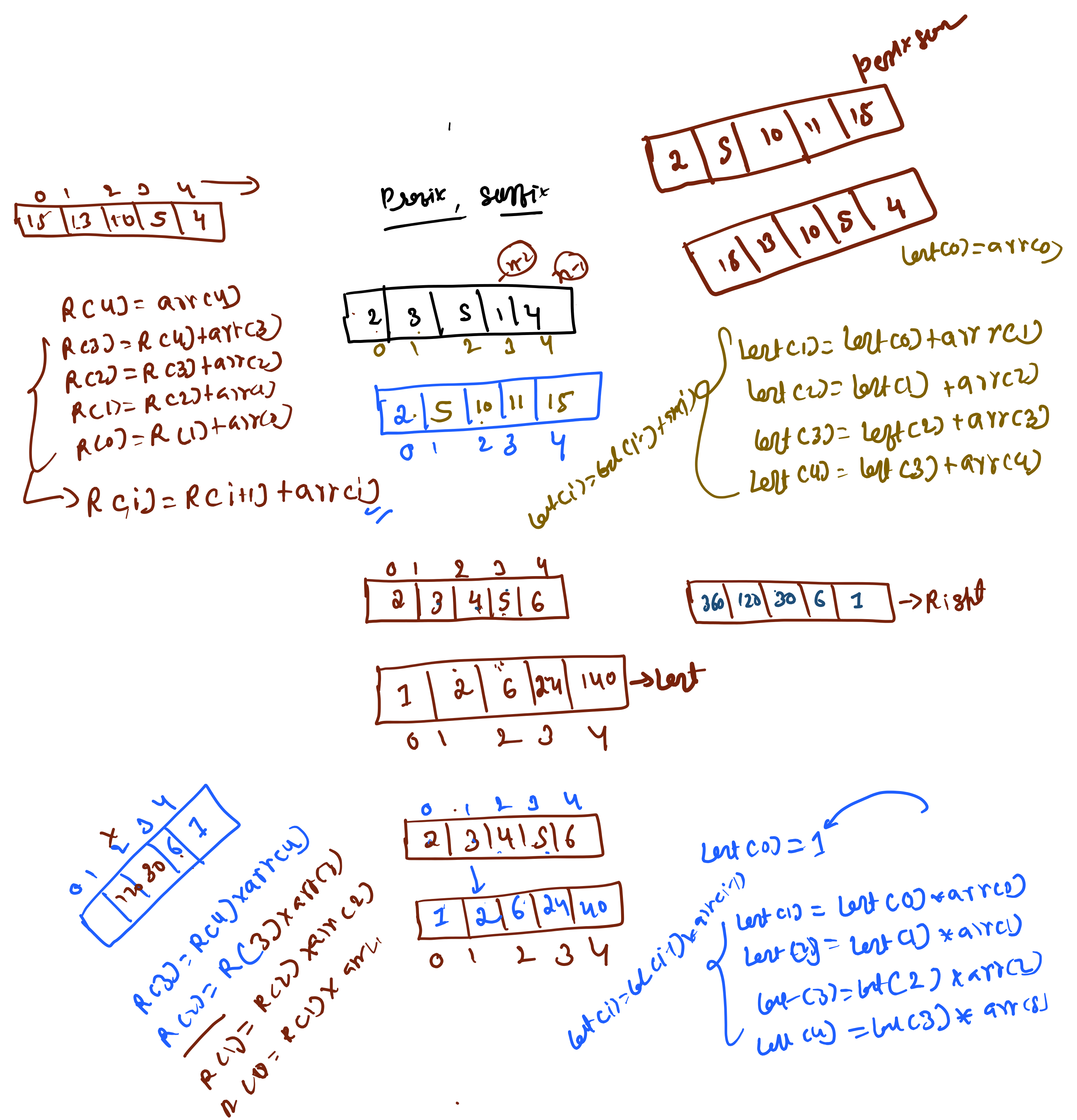
for(i=0 i<arr.length i++){
if(arr[i] > max){
max=arr[i]
}
}

i i i i j j j j j

{ 3, 5, 6, 2, 4, 16, 7, 8, 9 };

9 8 7 16 4 2 6 5 3

two Pointer



prefix sum
2 5 10 11 15
15 13 10 5 4
left[0]=arr[0]
left[1]= left[0]+arr[1]
left[2]= left[1]+arr[2]
left[3]= left[2]+arr[3]
left[4]= left[3]+arr[4]
left[i]=left[i-1]+arr[i]
0 1 2 3 4
15 13 10 5 4
R[4]= arr[4]
R[3]=R[4]+arr[3]
R[2]=R[3]+arr[2]
R[1]=R[2]+arr[1]
R[0]=R[1]+arr[0]
R[i]=R[i+1]+arr[i]
Prefix, Suffix
2 3 5 1 4
0 1 2 3 4
2 5 10 11 15
0 1 2 3 4
0 1 2 3 4
2 3 4 5 6
360 120 30 6 1 ->Right
1 2 6 24 140 ->Left
0 1 2 3 4
2 3 4 5 6
1 2 6 24 120
0 1 2 3 4
left[0]=1
left[1] = left[0]*arr[0]
left[2] = left[1]*arr[1]
left[3]=left[2]*arr[2]
left[4] =left[3]*arr[3]
left[i]=left[i-1]*arr[i-1]
0 1 2 3 4
120 30 6 1
R[3]=R[4]*arr[4]
R[2]=R[3]*arr[3]
R[1]=R[2]*arr[2]
R[0]=R[1]*arr[1]